INCH-POUND
MIL-C-11015/13F
25 October 2021
SUPERSEDING
MIL-C-11015/13E
14 May 2001

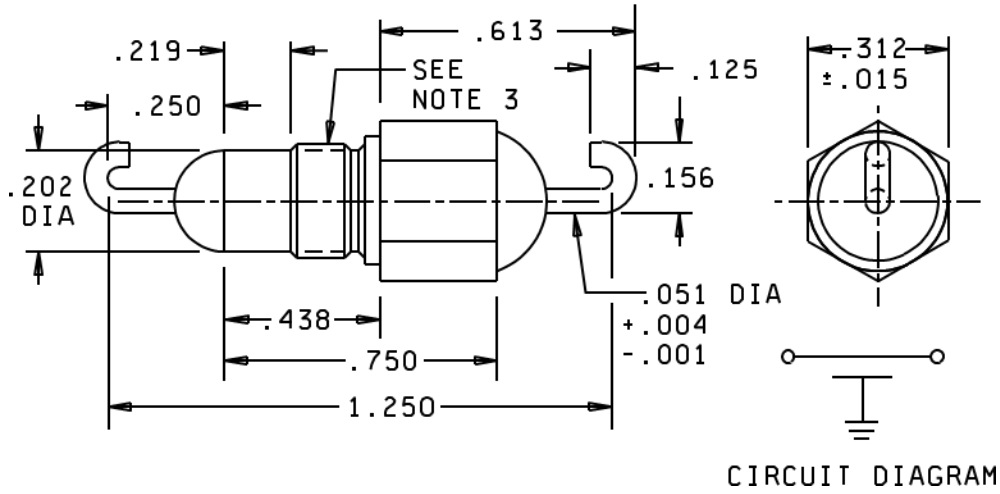
MILITARY SPECIFICATION SHEET

CAPACITOR, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), (FEED-THRU), STYLE CK70

INACTIVE FOR NEW DESIGN AFTER 31 MARCH 1999.
FOR REPLACEMENT PURPOSES ONLY

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-11015.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. Threaded portion shall be .250-28 UNF-2A and shall extend to within .047 inch (1.19 mm) of the shoulder.
4. A nut and internal-tooth lockwasher shall be supplied with each capacitor. The nut shall be .312 inch (7.92 mm) across flats and .125 inch (3.18 mm) thick. The lockwasher shall be silver plated bronze or brass, .4 inch (10.2 mm) outside diameter, .26 inch (6.6 mm) inside diameter.
5. Unless otherwise specified, tolerance is ± .031 inch (0.79 mm).

Inches	mm
.001	0.03
.004	0.10
.015	0.38
.051	1.30
.125	3.18
.156	3.96
.202	5.13
.219	5.56
.250	6.35
.312	7.92
.438	11.13
.613	15.57
.750	19.05
1.250	31.75

FIGURE 1. Dimensions and configuration.

AMSC N/A
FSC 5910

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REQUIREMENTS

Dimensions and configuration: See figure 1.

Case type: Feed thru.

Capacitance value: See table I.

Capacitance tolerance: ± 10 percent (K) or ± 20 percent (M) as shown in table I.

Rated temperature: -55°C to +85°C, voltage derated linearly to 50 percent at +125°C (see figure 2).

Dielectric withstanding voltage: In accordance with MIL-C-11015.

Dielectric (DWV):

Test voltage: 2,000 volts dc for 1,000 volt capacitors, or 250 percent of rated voltage for 500-volt capacitors, applied between the case and either terminal.

Body insulation: Not applicable.

TABLE I. Style CK70 characteristics.

Part or identifying number (PIN) 1/	Rated voltage (V_{dc})	Rated temperature and voltage-temperature limits	Capacitance (pF)	Capacitance tolerance
CK70AX100-	1,000	AX	10	K, M
CK70AX330-	1,000	AX	33	K, M
CK70AX470-	1,000	AX	47	K, M
CK70AX680-	1,000	AX	68	K, M
CK70AX101-	1,000	AX	100	K, M
CK70AX471-	500	AX	470	K, M
CK70AX681-	500	AX	680	K, M
CK70AW102M	500	AW	1,000	M
CK70AW152M	500	AW	1,500	M

1/ Where applicable, the complete PIN will include an additional symbol to indicate capacitance tolerance.

Barometric pressure (reduced): In accordance with MIL-C-11015 and MIL-STD-202-105, condition B (50,000 ft).

Test potential: 100 percent of rated voltage for 1,000-volt capacitors, or 150 percent of rated voltage for 500-volt capacitors, applied between the case and either terminal.

Insulation resistance (IR): In accordance with MIL-C-11015 and MIL-STD-202-302, condition B. 20,000 megohms, minimum, measured between either insulated point and the capacitor case.

Dissipation factor (DF): 1.5 percent, maximum (voltage-temperature limit X) or 2.0 percent, maximum (voltage-temperature limit W).

Vibration, high frequency: In accordance with MIL-C-11015 and MIL-STD-202-204, condition D (20 g's).

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Thermal shock and immersion: In accordance with MIL-C-11015.

IR: 3,000 megohms, minimum.

Terminal strength: Not applicable.

Moisture resistance: In accordance with MIL-C-11015 and MIL-STD-202-106.

DWV: 2,000 volts dc for 1,000-volt capacitors, or 250 percent of rated voltage for 500-volt capacitors.

IR: 3,000 megohms, minimum.

Cap.: Within tolerance of table I value.

Solderability: In accordance with MIL-C-11015; 2 terminals.

Life (at elevated ambient temperature): In accordance with MIL-C-11015.

Test potential: 200 percent of rated voltage for 500-volt capacitors, or 150 percent of rated voltage for 1,000-volt capacitors.

IR: 500 megohms, minimum (at +85°C) and 10,000 megohms, minimum (at +25°C).

DF: 3.0 percent, maximum (voltage-temperature limit X) or 4.0 percent, maximum (voltage-temperature limit W) (at +25°C).

Marking in accordance with MIL-C-11015.

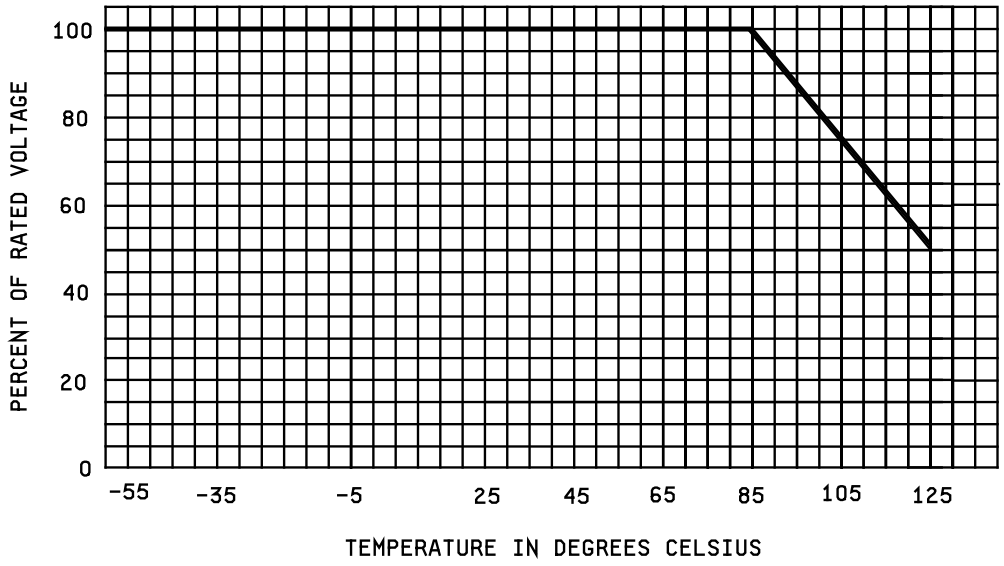


FIGURE 2. Voltage derating with temperature.

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Referenced documents: In addition to MIL-C-11015, this document references the following:

MIL-STD-202-105
MIL-STD-202-106
MIL-STD-202-204
MIL-STD-202-302

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
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